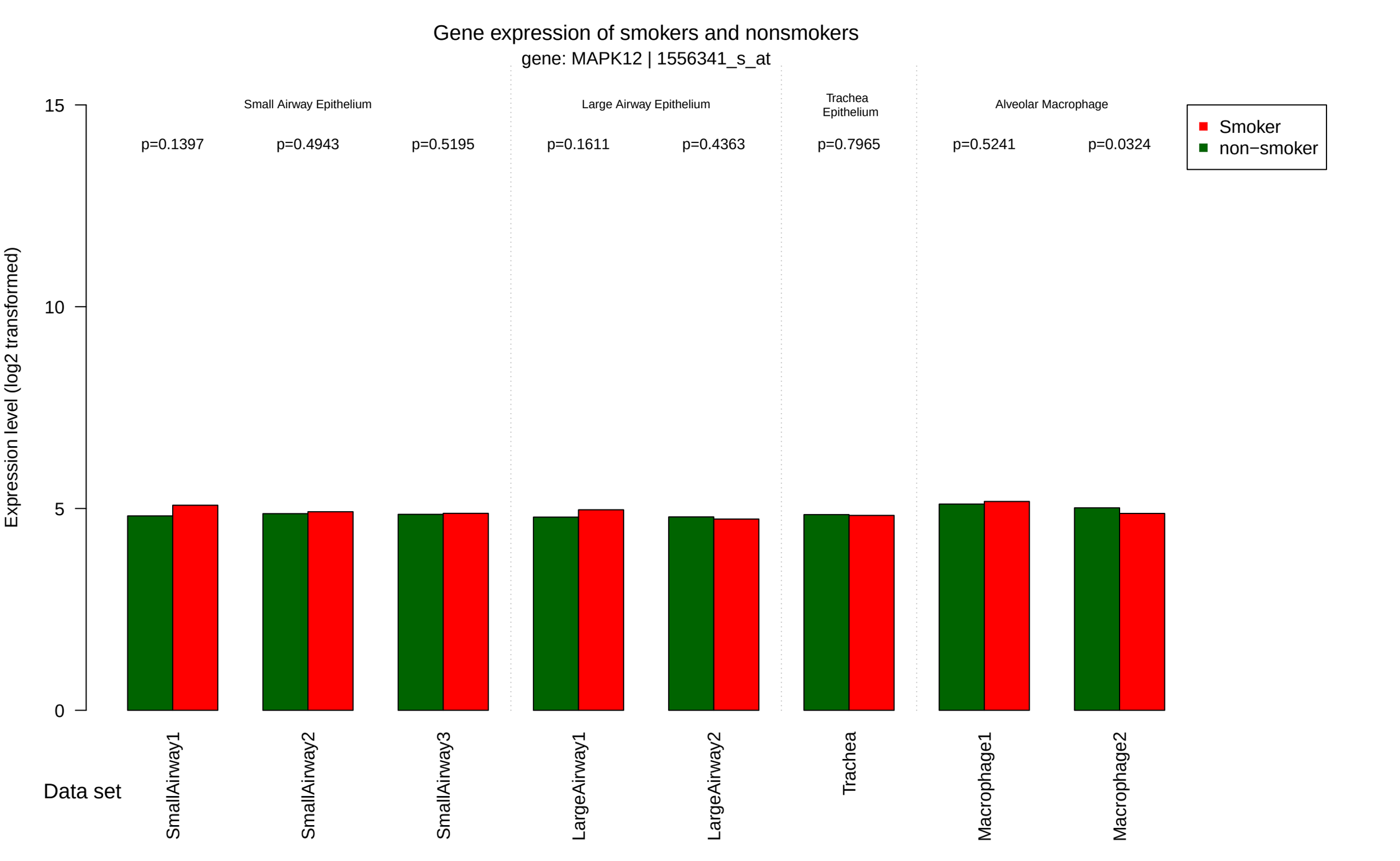
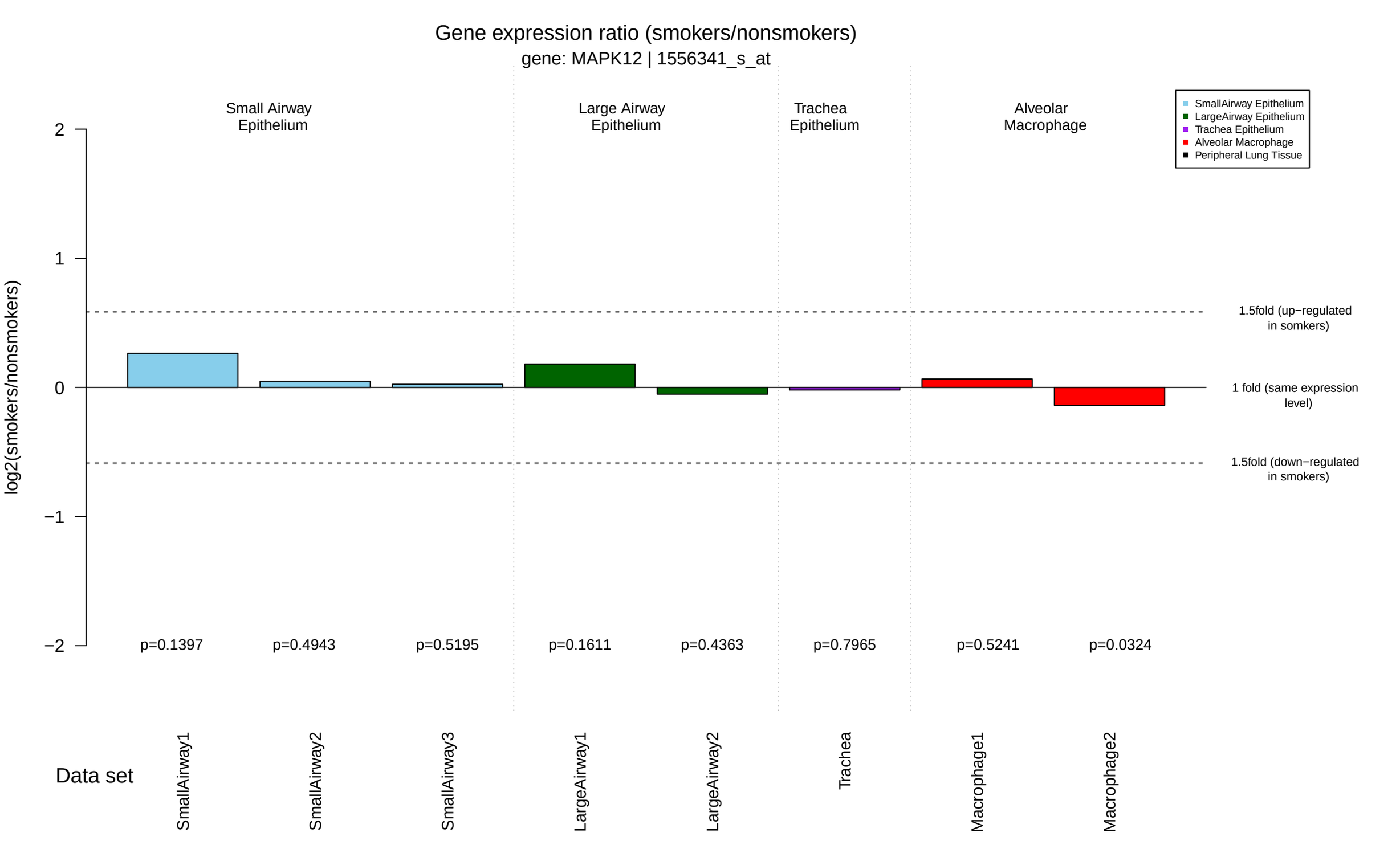
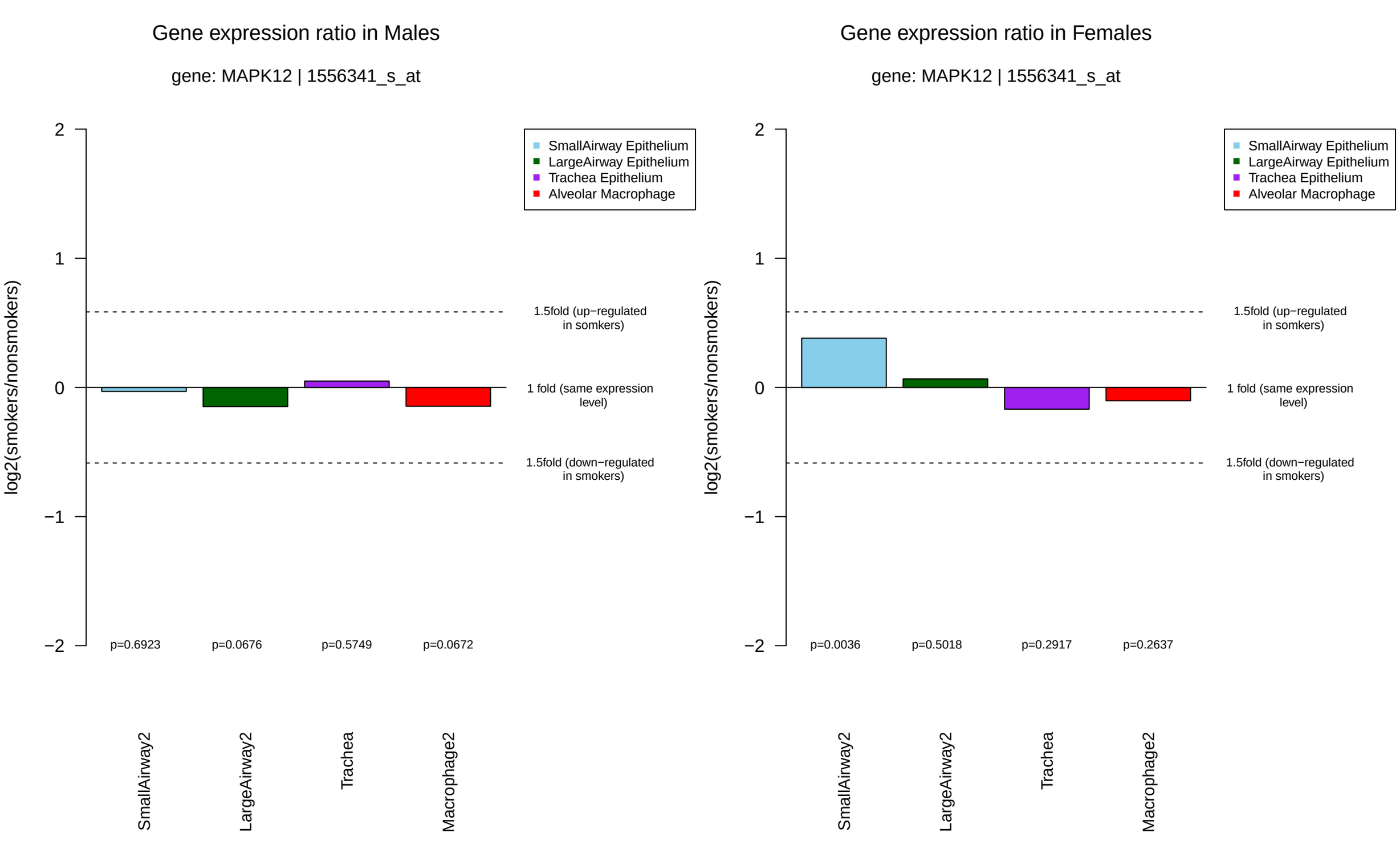
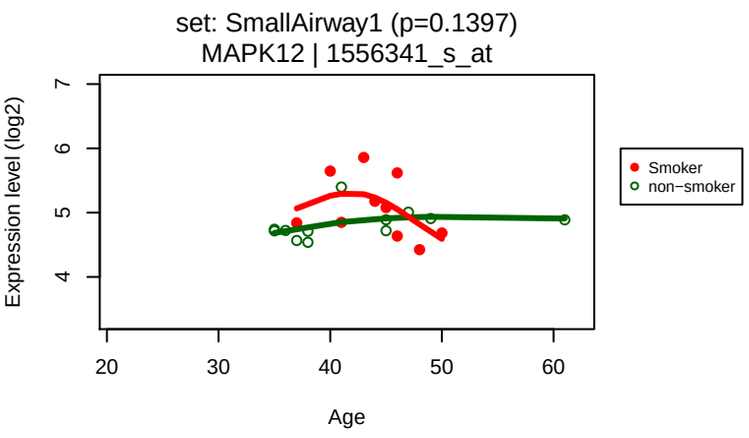
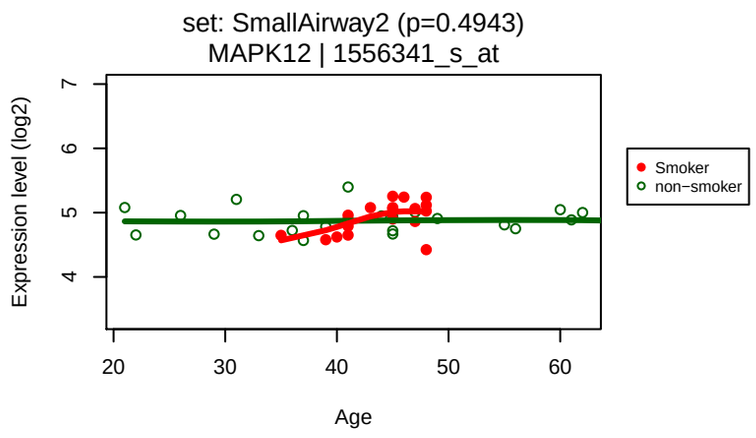
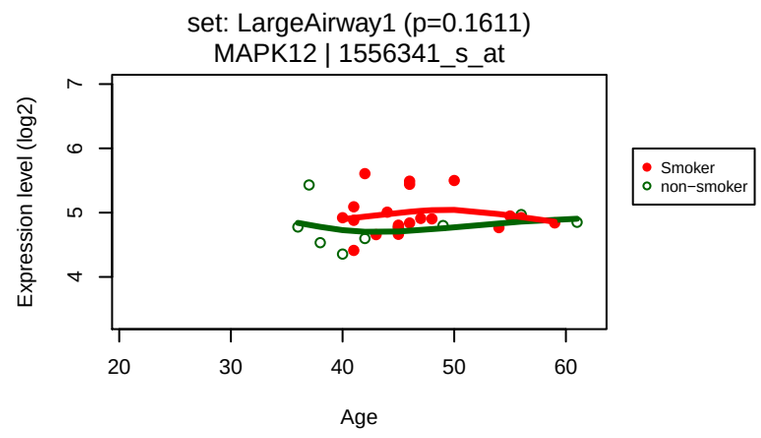

Gene expression of smokers and nonsmokers
gene: MAPK12 | 1556341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1397
p=0.4943
p=0.5195
p=0.1611
p=0.4363
p=0.7965
p=0.5241
p=0.0324
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

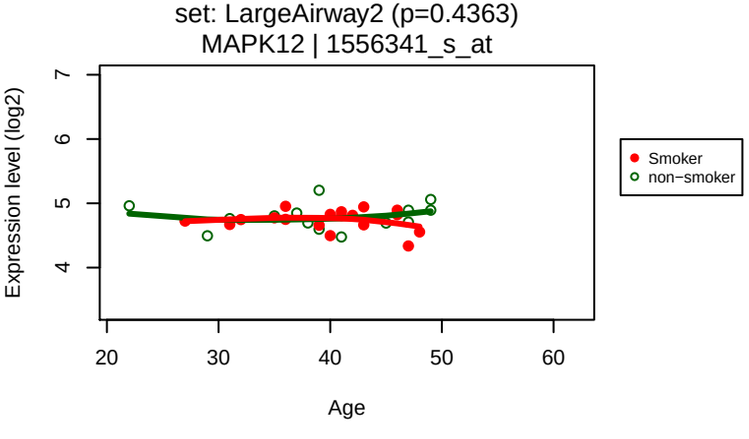

Gene expression ratio (smokers/nonsmokers)
gene: MAPK12 | 1556341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1397
p=0.4943
p=0.5195
p=0.1611
p=0.4363
p=0.7965
p=0.5241
p=0.0324
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

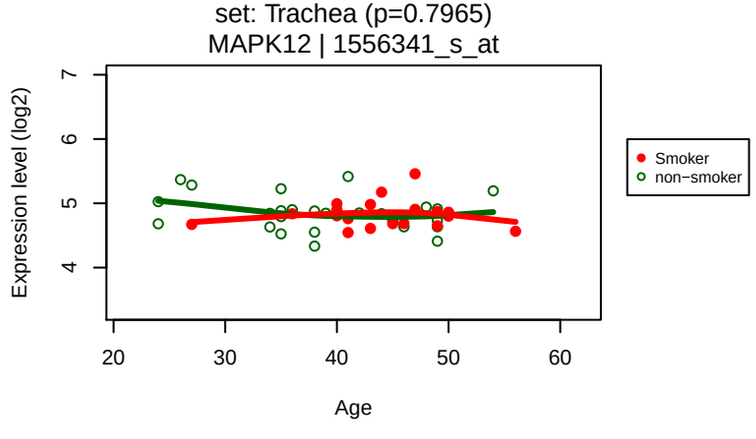

Gene expression ratio in Males
gene: MAPK12 | 1556341_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6923
p=0.0676
p=0.5749
p=0.0672
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK12 | 1556341_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0036
p=0.5018
p=0.2917
p=0.2637
SmallAirway2
LargeAirway2
Trachea
Macrophage2

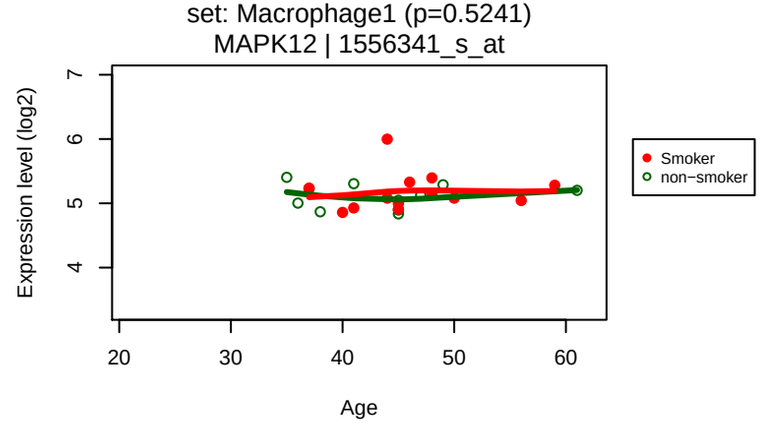

set: SmallAirway1 (p=0.1397)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker

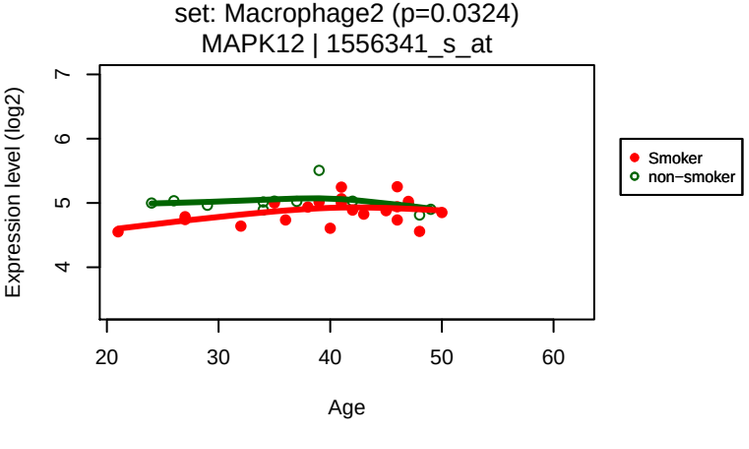

set: SmallAirway2 (p=0.4943)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker

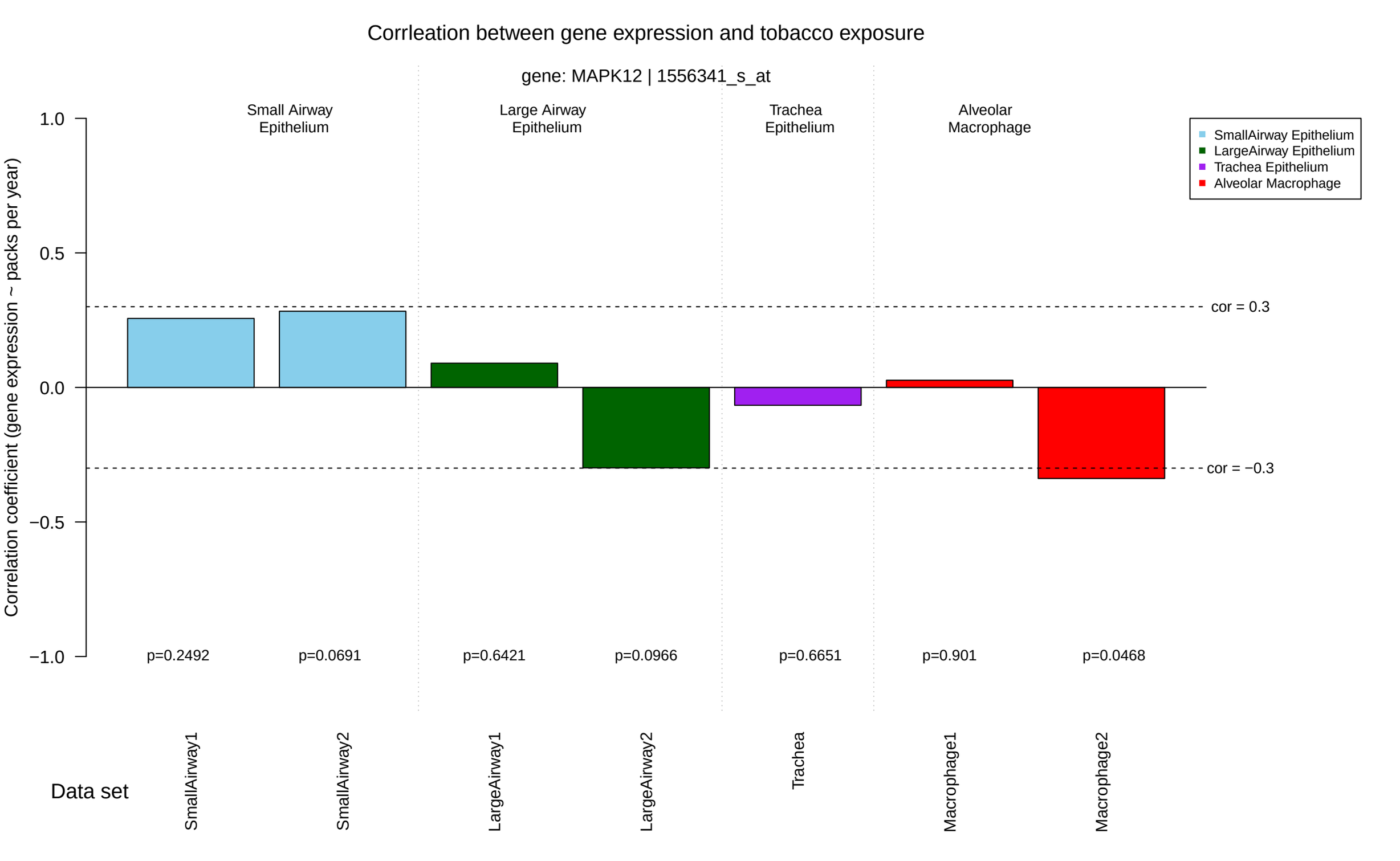

set: LargeAirway1 (p=0.1611)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4363)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7965)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5241)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0324)
MAPK12 | 1556341_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK12 | 1556341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2492
p=0.0691
p=0.6421
p=0.0966
p=0.6651
p=0.901
p=0.0468
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2